

$$\begin{array}{ccc}
 f & \xrightarrow{H} & Hf \\
 T_\tau \downarrow & & \downarrow T_\tau \\
 T_\tau f & \xrightarrow{H} & T_\tau(Hf)
 \end{array}$$

$$T_\tau f(x) = f(x - \tau)$$